

SAMUEL XIFARAS

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New York, NY

SUMMARY

Full stack software engineer with 5+ years shipping production systems at Stripe, Paperless Parts, and an early-stage incubator, across backend services, search infrastructure, and customer-facing product work. Graduate research background in formal methods, with published work in mechanized reasoning. Bring proof-level rigor to system design while staying focused on users and product outcomes.

SKILLS

Languages: Python, Ruby, JavaScript/TypeScript, Java, C#, C, SQL, HTML/CSS, Racket (Scheme), Common Lisp
Technologies and frameworks: Django, Flask, FastAPI, ReactJS, Docker, PostgreSQL, MongoDB, Terraform, OpenAI Agents SDK, Cursor, Jupyter, ACL2/ACL2s, Lean
Cloud services: AWS (EC2, S3, ECS, Route53, RDS, Managed OpenSearch, SQS, SNS, Lambda), Datadog, Prometheus, Grafana, Logscale, Splunk

WORK EXPERIENCE

Stripe

Software Engineer

New York, NY

June 2025 – Present

- Led the technical design that cut an endpoint on a time-interval-based configuration store from **30s timeouts to <10ms** via interval-aware indexing; specified it clearly enough to delegate during an absence
- Shipped the revenue projection backend for **Revenue Share** deals, powering some of Stripe's largest platform partnerships
- Drove design clarity on a customer-facing fee schedule UI using support-interaction data and an interactive prototype to align product, design, and engineering
- Mentored a summer intern on project prioritization and career growth

Raven (RA Ventures)

Full Stack Software Engineer III

Boston, MA

February 2025 – June 2025

- **Founding engineer** on the first incubated company build
- Built **two AI-native prototypes**: a diagnostic chatbot and a clinical trial compensation calculator
- Built **evaluation metrics** and **test suites** to monitor the performance of an LLM-based system

Paperless Parts

Software Engineer II (promoted from I)

Boston, MA

August 2021 – January 2025

- Built an RFQ forwarding service processing roughly **400 emails/day**; its top 10 users represent **\$219k** in ARR
- Delivered the encryption standards required for **FedRAMP Moderate compliance**
- Led a latency investigation after production incidents and cut reindexing on common updates by **10x** by denormalizing **OpenSearch** indices
- Designed a greenfield faceted search backend modeling **10 interdependent facets** whose availability depends on upstream selections, using **functional programming techniques** for tight state control
- Led a backend Community of Practice of **10 engineers** for eight months, cutting flaky tests from about **20 to under 5** via a test-reliability initiative
- Mentored **two** co-ops

Paperless Parts

Software Engineering and DevOps Co-op

Boston, MA

January 2021 – May 2021

- Shipped full-stack workflow features in the core product with **ReactJS** and **Django**
- **Worked directly with a customer** to build a Paperless Parts-ERP integration

ference

Data Science/Software Engineering Co-op

Cambridge, MA

September 2019 – August 2020

- Led frontend development with **ReactJS** for a "Competitive Landscape" data mining tool for Janssen
- **Interfaced directly with Janssen clients** at monthly meetings on progress and feature requests
- Built **2 APIs**—for storing image annotation data and retrieving financial market data—using **Python** and **MongoDB**
- Produced biomedical data analyses and visualizations with **Pandas**, **scikit-learn**, and **R** for **2** Janssen client projects

RESEARCH EXPERIENCE

Apart Research <i>Research Fellow, Secure Program Synthesis Fellowship</i>	Remote July 2026 – Present
· Built a functional Ruby semantics in Lean passing 992 of 1,304 CRuby bootstrap tests with byte-exact output · Designing theorems and proof decomposition for a type soundness result on a subset of Sorbet; LLM agents fill in proof bodies	
Khoury College of Computer Sciences, Northeastern University <i>Master's Thesis</i>	Boston, MA January 2023 – November 2024
Thesis Title: Leveraging Type Annotations for Effective Fuzzing of Python Programs Thesis Advisor: Dr. Panagiotis Manolios PDF Link: samx.io/papers/thesis.pdf Synopsis: Static analysis tooling for Python emits many false positives. My advisor and I hypothesized that a hybrid static-dynamic approach would work better for a dynamic, duck-typed language like Python, so I built a fuzzer that extracts Python type annotations and uses them to fuzz programs more intelligently. I evaluated it on open-source repositories, finding and reporting several real bugs. · Implemented a fuzzing tool from scratch in Python with strong object-oriented design principles to facilitate rapid extension · Evaluated across 15 open-source repositories; reported 3 real-world bugs, all accepted and patched by maintainers · Built infrastructure to orchestrate experiment runs and results collection on a SLURM-based HPC environment · Dockerized the application, in preparation for potential commercial deployment	
Khoury College of Computer Sciences, Northeastern University <i>Research Assistant</i>	Boston, MA January 2022 – May 2022
· Implemented a neural network portfolio—an ensemble of several neural networks—with PyTorch · Used open-source neural network verification tooling to formally prove bounded error for a learned database index · Built infrastructure to run experiments and collect results	

PUBLICATIONS

Samuel Xifaras , Panagiotis Manolios, Andrew Walter, and William Robertson. <i>An Enumerative Embedding of the Python Type System in ACL2s</i> . Submitted to ACL2 Workshop 2025, Accepted March 9, 2025. https://arxiv.org/pdf/2507.19015 .
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EDUCATION

Khoury College of Computer Sciences, Northeastern University MS in Computer Science GPA: 4.00/4.00 Relevant Coursework: Computer-Aided Reasoning; Compilers Thesis Title: Leveraging Type Annotations for Effective Fuzzing of Python Programs Thesis Advisor: Dr. Panagiotis Manolios	Boston, MA September 2022 – December 2024
Khoury College of Computer Sciences, Northeastern University BS in Computer Science GPA: 3.92/4.00 Honors: Khoury College Dean's List, 6 semesters Positions: Teaching Assistant, Object Oriented Design	Boston, MA September 2018 – May 2022
Phillips Academy Andover GPA: 5.43/6.00	Andover, MA September 2014 – June 2018

COMMUNITY ENGAGEMENT

Volunteer gardener , The Family Pantry of Cape Cod, Harwich, MA	Summer 2022 – Present
Interests: gardening, hiking, biking, tennis, electronic music production, investing, classical piano	